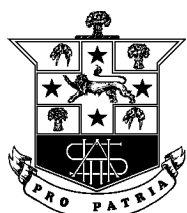


STUDENT NAME: _____

TEACHER: _____



**HURLSTONE
AGRICULTURAL
HIGH
SCHOOL**

2019

**HIGHER
SCHOOL
CERTIFICATE
HSC Task 4 – Trial HSC**

Mathematics

Examiners

- Mr D Potaczala Ms P Biczko, Ms. T Tarannum, Ms D Crancher, Mr J Dillon

**General
Instructions**

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen
- NESA-approved calculators may be used
- A Reference sheet is provided
- In Questions 11 – 16, show relevant mathematical reasoning and/or calculations

**Total marks:
100****Section I – 10 marks (pages 3 – 8)**

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

Section II – 90 marks (pages 9 – 17)

- Attempt Questions 11 – 16
- Allow about 2 hours and 45 minutes for this section

SECTION I

10 marks

Attempt Questions 1 – 10

Allow about 10 minutes for this sections.

Use the multiple-choice answer sheet for Questions 1 – 10

Question 1

What is the value of $\frac{e^2}{6}$, correct to 3 significant figures?

- A. 1.231
- B. 1.232
- C. 1.23
- D. 1.22

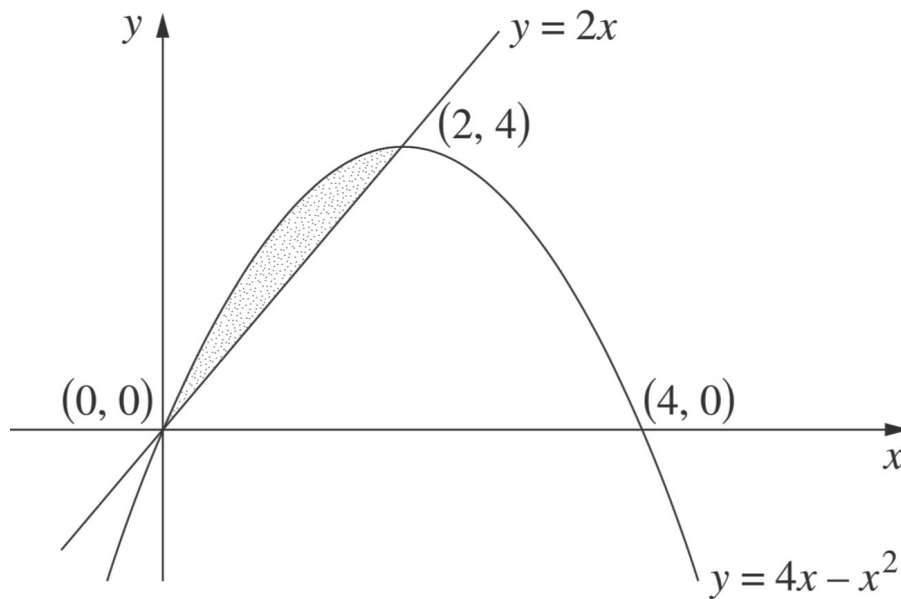
Question 2

What is the solution to the equation $\cos 2x = \frac{1}{2}$ in the domain $-\pi \leq x \leq \pi$?

- A. $x = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{-5\pi}{6}, \frac{-\pi}{6}$
- B. $x = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$
- C. $x = \frac{\pi}{12}, \frac{11\pi}{12}, \frac{13\pi}{12}, \frac{23\pi}{12}$
- D. $x = \frac{\pi}{12}, \frac{11\pi}{12}, \frac{-11\pi}{12}, \frac{-\pi}{12}$

Question 3

The diagram shows the parabola $y = 4x - x^2$ meeting the $y = 2x$ line at $(0, 0)$ and $(2, 4)$.

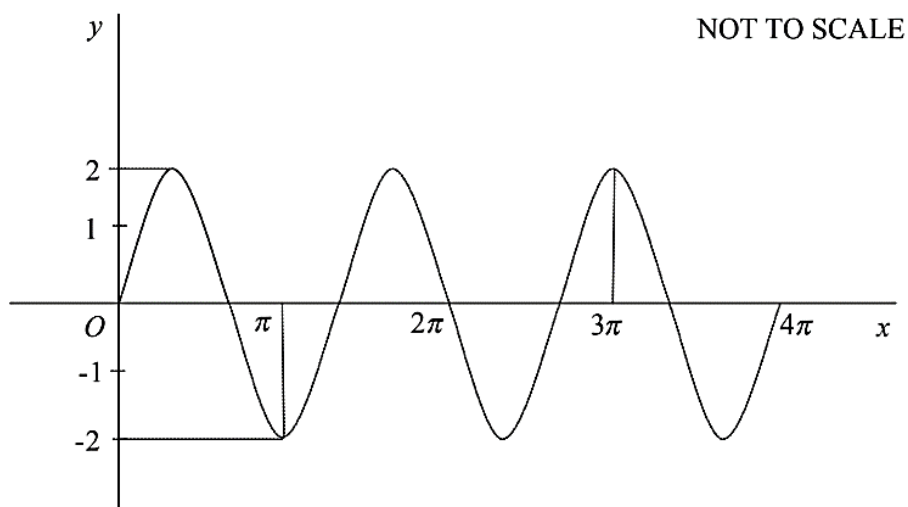


Which expression gives the area of the shaded region bounded by the parabola and the line?

- A. $\int_0^4 x^2 - 2x \, dx$
- B. $\int_0^4 2x - x^2 \, dx$
- C. $\int_0^2 x^2 - 2x \, dx$
- D. $\int_0^2 2x - x^2 \, dx$

Question 4

Which trigonometric equation represents the graph given below?



- A. $y = 2 \sin\left(\frac{2\pi}{3}x\right)$
- B. $y = 2 \sin\left(\frac{3\pi}{2}x\right)$
- C. $y = 2 \sin\left(\frac{2}{3}x\right)$
- D. $y = 2 \sin\left(\frac{3}{2}x\right)$

Question 5

The derivative of the curve $f(x) = 4^x$ is given by which of the following?

- A. $f'(x) = 4 \times 4^x$
- B. $f'(x) = \log_e 4 \times 4^x$
- C. $f'(x) = 4 \times e^x$
- D. $f'(x) = \log_e 4 \times e^x$

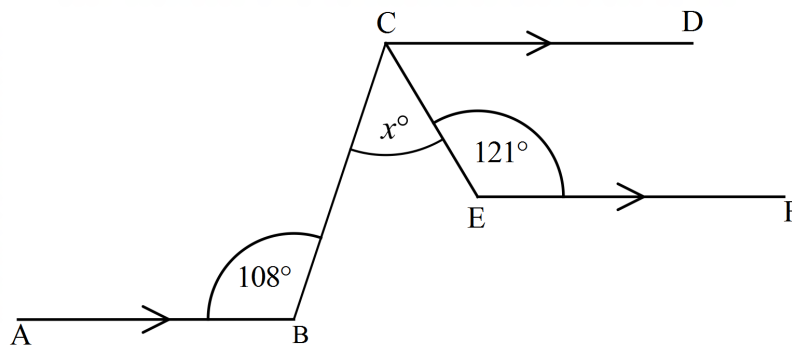
Question 6

$\int e^{4x+1} dx$ is given by which of the following?

- A. $\frac{e^{4x+1}}{\ln 4} + c$
- B. $\frac{e^{4x+1}}{4} + c$
- C. $e^{4x+1} + c$
- D. $4e^{4x+1} + c$

Question 7

In the diagram below, what is the value of x ?



- A. 49°
- B. 59°
- C. 108°
- D. 121°

Question 8

In a class of 30 students, 8 have brown hair, 12 have brown eyes and 4 have both.

What is the probability that a student selected at random has either brown hair or brown eyes but not both?

- A. $\frac{2}{15}$
- B. $\frac{2}{5}$
- C. $\frac{8}{15}$
- D. $\frac{2}{3}$

Question 9

A flat circular disc is being heated so that the rate of increase of the area (A in m^2), after t hours, given by $\frac{dA}{dt} = \frac{1}{8}\pi t$. Initially the disc has a radius of 2 metres.

Which of the following is the correct expression for the area after t hours?

- A. $A = \frac{1}{8}\pi t^2$
- B. $A = \frac{1}{16}\pi t^2$
- C. $A = \frac{1}{8}\pi t^2 + 4\pi$
- D. $A = \frac{1}{16}\pi t^2 + 4\pi$

Question 10

It is assumed that the number $N(t)$ of ants in a certain nest at time $t \geq 0$ is given by

$$N(t) = \frac{A}{1 + e^{-t}}$$

where A is a constant and t is measured in months.

At time $t = 0$, $N(t)$ is estimated at 2×10^5 ants. What is the value of A ?

- A. 2×10^5
- B. 2×10^{-5}
- C. 4×10^5
- D. 4×10^{-5}

SECTION II

90 marks

Attempt Questions 11 – 16

Allow about 2hour 45 minutes for this section

Answer each question in a **SEPARATE** writing booklet. Additional writing booklets are available.

In Questions 11 – 16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Start a new booklet

(a) Solve $2^{2x+1} = 32$. 2

(b) Solve $|x - 2| \leq 3$. 2

(c) Factorise fully $36 - x^2$. 2

(d) A parabola has a directrix $y = -2$ and a focus $(5, 4)$. Find the coordinates of the vertex. 2

(e) (i) Rationalise the denominator in the expression: 2

$$\frac{1}{\sqrt{n} + \sqrt{n+1}}$$

where n is an integer and $n \geq 1$.

(ii) Using your result from part (i), or otherwise, find the value of the sum: 2

$$\frac{1}{\sqrt{1} + \sqrt{2}} + \frac{1}{\sqrt{2} + \sqrt{3}} + \frac{1}{\sqrt{3} + \sqrt{4}} + \dots + \frac{1}{\sqrt{99} + \sqrt{100}}$$

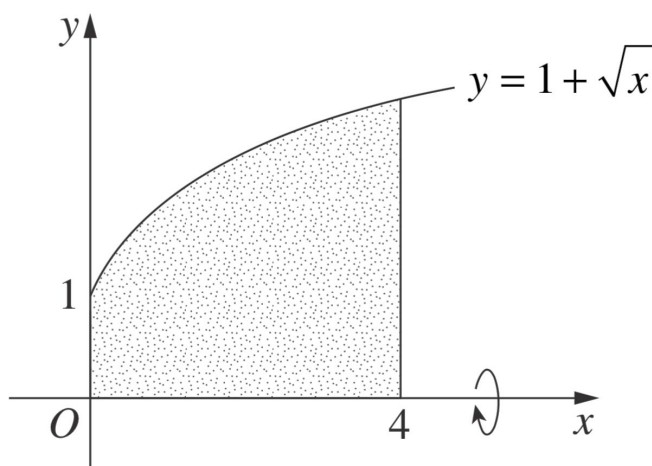
(f) A point $P(x, y)$ moves so that the sum of the squares of the distances from each of the points $A(-1, 0)$ and $B(3, 0)$ is equal to 40. 3
Show that the locus of $P(x, y)$ is a circle and state its radius and centre.

Question 12 (15 marks) Start a new booklet

(a) Evaluate $\lim_{x \rightarrow 2} \frac{x^3 - 8}{x - 2}$. 2

(b) Differentiate $y = \frac{x-1}{3x^2}$. 2

- (c) The region bounded by the $y = 1 + \sqrt{x}$ and the x -axis between $x = 0$ and $x = 4$ is rotated about the x -axis to form a solid. Find the volume of the solid. 3



- (d) Consider the curve $y = x^3 - x^2 - x + 1$.
- (i) Find the stationary points and determine their nature. 4
- (ii) Given the point $P\left(\frac{1}{3}, \frac{16}{27}\right)$ lies on the curve, prove that it is a point of inflection. 2
- (iii) Sketch the curve labelling the stationary points, point of inflection and y -intercept. 2

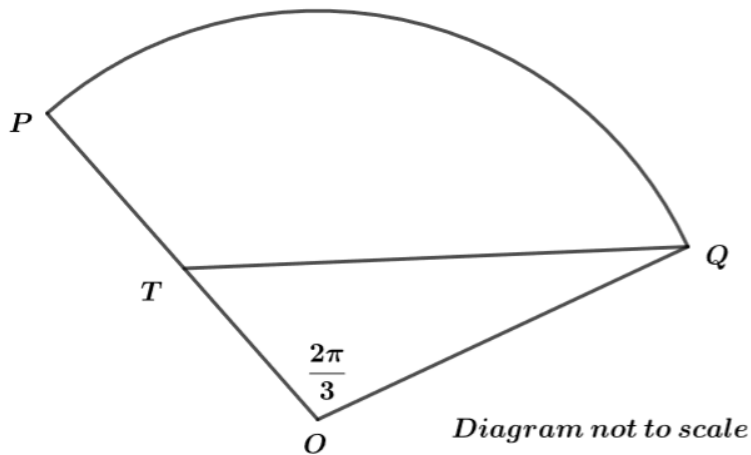
Question 13 (15 marks) Start a new booklet

- (a) Solve $\tan x = \frac{1}{3}$, where $0 \leq x \leq 2\pi$. **2**

Give your answer(s) in radians to two decimal places.

- (b) In the diagram, PQ is an arc of a circle with radius 10 cm and centre O . **3**

T is the midpoint of OP . Angle POQ is $\frac{2\pi}{3}$.

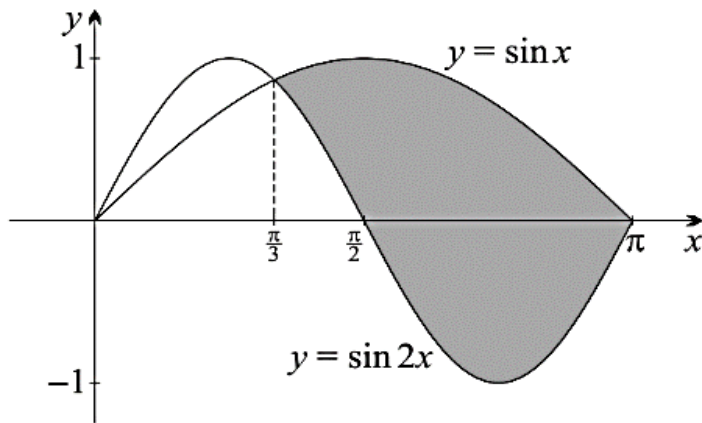


Find the perimeter of the shape PTQ in exact form.

Question 13 continues on next page

(c)

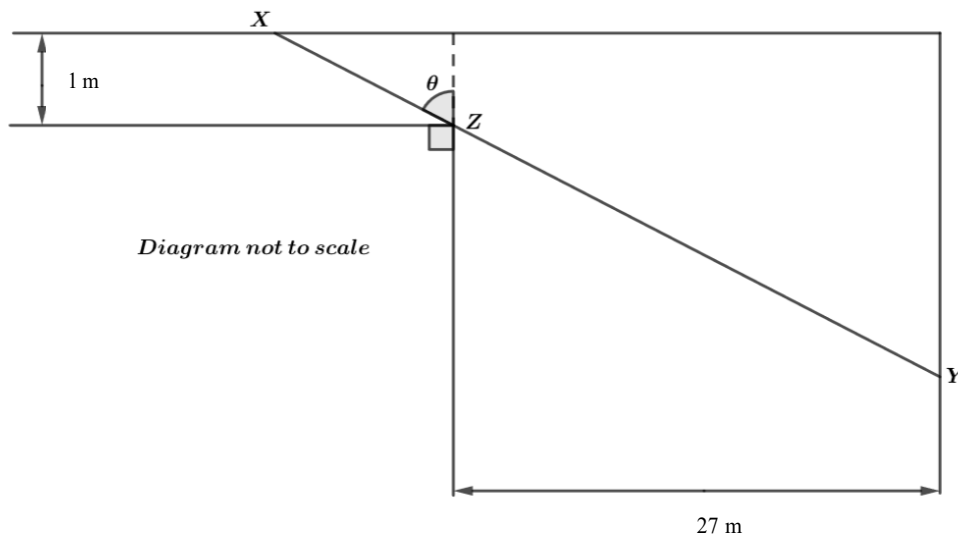
3



The diagram above shows the curves $y = \sin x$ and $y = \sin 2x$ for $0 \leq x \leq \pi$, intersecting at $x = 0$, $x = \frac{\pi}{3}$ and $x = \pi$.

Find the exact area of the shaded region bounded by the two curves.

- (d) A corridor of width 1 m enters a room of width 27 m, intersecting in a right angle, as shown in the diagram. A straight rope XY touches the corner Z as shown in the diagram below. Let L be the length of XY .



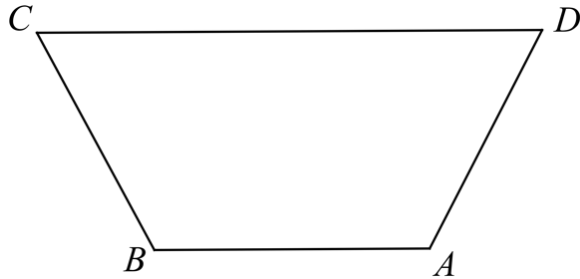
- (i) Show that $L = \frac{1}{\cos \theta} + \frac{27}{\sin \theta}$. 1
- (ii) Show that $\frac{dL}{d\theta} = \frac{\cos \theta}{\sin^2 \theta} (\tan^3 \theta - 27)$. 3
- (iii) Given that $0 < \theta < \frac{\pi}{2}$, find the minimum value of L . 3
- Give your answer as an exact value.

Question 14 (15 marks) Start a new booklet

- (a) Solve for k : $\log_5 8 = \log_5 k - 2 \log_5 3$. 2
- (b) Use the trapezoidal rule, with five function values to find the area bounded by the curve $y = \ln(x^2 - 1)$, the x axis and the lines $x = 2$ and $x = 4$. Give your answer to 2 decimal places. 3
- (c) Find $\int \frac{1}{2x-1} dx$ 2
- (d) Given $f(x) = \log_e \sqrt{2-x}$, find using the log laws, or otherwise, $f'(x)$. 2
- (e) (i) Sketch the curve $y = \log_e(x-1)$ and shade the area bounded by the curve, the x axis, the y axis and the line $y = \log_e 5$. 1
- (ii) Calculate the shaded area, giving your answer in simplest exact form. 3
- (f) (i) Show that $\frac{d}{dx}(xe^x) = xe^x + e^x$. 1
- (ii) Hence, find $\int xe^x dx$. 1

Question 15 (15 marks) Start a new booklet

- (a) $ABCD$ is a quadrilateral with $\angle ABC = \angle BAD$ and $BC = AD$.



Not to scale

- (i) Prove that $\triangle ABC \equiv \triangle BAD$. 2
- (ii) Why are $\angle CAB$ and $\angle ABD$ equal? 1
- (iii) Prove that $\angle DBC = \angle CAD$. 2
- (b) A triangle has vertices $A(1, -3)$, $B(3, 3)$ and $C(-3, 1)$.
- (i) Find the coordinates of L and M , the midpoints of AB and BC respectively. 1
- (ii) Show that LM is parallel to AC and that $LM = \frac{1}{2} AC$. 2
- (c) The senior students at a school decide to send a delegation of two to a conference.
The delegation will have one student from Year 11 and one from Year 12.
The candidates from Year 11 are Petra, Quentin and Rufus, who have probabilities of $\frac{1}{6}$, $\frac{1}{3}$ and $\frac{1}{2}$ respectively of being chosen by their classmates.
The candidates from Year 12 are Amelia, Bella, Charles and Diana, who have probabilities of $\frac{1}{8}$, $\frac{1}{4}$, $\frac{3}{8}$ and $\frac{1}{4}$ respectively of being chosen.
- (i) Draw a tree diagram to show the possible pairs in the delegation. Include the probability on each branch. 2
- (ii) What is the probability that the delegation includes either Quentin or Bella but not both? 1

Question 15 continues on next page

- (d) A worker invests $\$P$ at the beginning of each month into a retirement fund that pays 6% p.a. compounded monthly, on the money invested, for 20 years.
- (i) Show that after 2 months there is $\$P(1.005^2 + 1.005)$ in the fund. **1**
- (ii) Write an expression for the amount in the fund after one year. **1**
- (iii) The worker wishes to retire at the end of the 20 years with a lump sum of \$450 000. **2**
- What investment must the worker make at the beginning of each month to achieve this?

Question 16 (15 marks) Start a new booklet

- (a) A water tank has an initial capacity of 3000 litres and is leaking according to the formula $V = V_0 e^{-kt}$, where t is in hours.

(i) Show that $\frac{dV}{dt} = -kV$. 1

- (ii) What is the value of k if after 3 hours the volume is 2000 litres? 2
Give your answer correct to 3 decimal places.

- (iii) How long will it take for the amount of water in the tank to fall to 250 litres? 2
Give your answer correct to the nearest minute.

- (b) The rate of emission of carbon pollution C , in tonnes per year from a factory from 1st January 2011 is given by:

$$C = 500 - \left(\frac{10}{1+t} \right)^2 \text{ where } t \text{ is the time in years.}$$

- (i) What was the rate of emission of carbon pollution C on 1st January 2011? 1
- (ii) What was the rate of emission of carbon pollution C on 1st January 2013? 1
- (iii) What value does C approach as time passes? 1
- (iv) Draw a sketch of C as a function of t . 1
- (v) Calculate the total amount of carbon pollution emitted from the factory from 1st January 2011 to 1st January 2017? Answer correct to the nearest tonne. 2

Question 16 continues on next page

- (c) A factory produces mobile phones. The annual production of phones, M , at time t years, is given by:

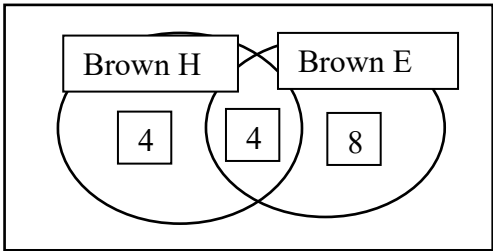
$$M = 2000e^{kt} \text{ where } k \text{ is a constant.}$$

After five years, the production has increased to 3200 phones per annum.

- | | |
|---|----------|
| (i) Find the value of k . | 1 |
| (ii) What is the predicted production after 10 years? | 1 |
| (iii) How many years will it take for the production to double its original output? | 1 |
| (iv) Find the rate of increase in production when the factory has been operating for 5 years. | 1 |

End of Paper

Hurlstone Agricultural High School Year 12 Trial Higher School Certificate Question No. 11		Mathematics Solutions and Marking Guidelines	Examination 2019
Outcomes Addressed in this Question			
PE2	Uses multi-step deductive reasoning in a variety of contexts		
PE6	Makes comprehensive use of mathematical language, diagrams & notation for communicating in a wide variety of situations		
HE1	Appreciates interrelationships between ideas drawn from different areas of mathematics		
HE7	evaluates mathematical solutions to problems and communicates them in an appropriate form		
Outcome	Solutions	Marking Guidelines	
PE2	Q1. $\frac{e^2}{6} = 1.2315 \dots$ 3 significant figures is 1.23 .	1 mark	
PE2	Q2. $\cos 2x = \frac{1}{2}$ in the domain $-\pi \leq x \leq \pi$. Since it is $2x$, change the domain to $-2\pi \leq 2x \leq 2\pi$. $2x = \cos^{-1}\left(\frac{1}{2}\right)$ $2x = -\frac{5\pi}{3}, -\frac{\pi}{3}, \frac{\pi}{3}, \frac{5\pi}{3}$ $x = -\frac{5\pi}{6}, -\frac{\pi}{6}, \frac{\pi}{6}, \frac{5\pi}{6}$	1 mark	
PE2	Q3. <i>The parabola $y = 4x - x^2$ is above the line $y = 2x$.</i> $\int_0^2 4x - x^2 - 2x \, dx$ $= \int_0^2 2x - x^2 \, dx$	1 mark	
PE2	Q4. The graph has, amplitude=2 and period = $\frac{3}{2}$ $\therefore y = 2\sin\left(\frac{3}{2}x\right)$	1 mark	
PE6	Q5. $\frac{d}{dx}(a^x) = a^x \ln a$ $f(x) = 4^x$ $f'(x) = 4^x \ln 4$ $f'(x) = 4^x \times \log_e 4$	1 mark	

PE6	Q6. $\int(e^{f(x)}) = \frac{1}{f'(x)}e^{f(x)} + c$ $\int e^{4x+1} dx$ $f(x) = 4x + 1 \text{ and } f'(x) = 4$ $\therefore \frac{e^{4x+1}}{4} + c$	1 mark
PE6	Q7. $\angle ABC = 108^\circ \text{ (given)}$ $\angle BCD = 108^\circ \text{ (alternative angles on parallel lines)}$ $\angle CEF = 121^\circ \text{ (given)}$ $\angle ECD = 180^\circ - 121^\circ = 59^\circ \text{ (cointerior angles on parallel lines)}$ $\angle BCD - \angle ECD = \angle BCE$ $\therefore 108^\circ - 59^\circ = \mathbf{49^\circ}$	1 mark
PE2	Q8  There are 4 students who only have brown hair and 8 students who only have brown eyes out of the 30 students. $\therefore \frac{12}{30} = \frac{2}{5}$	1 mark
PE2	Q9. $\frac{dA}{dt} = \frac{1}{8}\pi t$ $A = \int \frac{1}{8}\pi t dt$ $= \frac{1}{8} \times \frac{1}{2} \times \pi \times t^2 + c$ To find the area of a circle, we can replace $A = \pi r^2$ $\pi r^2 = \frac{1}{16}\pi t^2 + c$ At $t=0$, $r=2$ $4\pi = 0 + c$ $c = 4\pi$ $\therefore A = \frac{1}{16}\pi t^2 + 4\pi$	1 mark

HE7, HE1	Q10. At time $t = 0$, $N(t) = \frac{A}{1+e^{-0}} = \frac{A}{1+1} = \frac{A}{2}$ $N(t)$ is estimated at 2×10^5. $2 \times 10^5 = \frac{A}{2}$ $A = 2 \times 2 \times 10^5$ $A = 4 \times 10^5$	1 mark
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Year 12 Mathematics Trial 2019		
Question No. 11		Solutions and Marking Guidelines
Outcomes Addressed in this Question		
P3 - performs routine arithmetic and algebraic manipulation involving surds, simple rational expressions and trigonometric identities P4 - chooses and applies appropriate arithmetic, algebraic, graphical and geometric techniques		
Outcome	Solutions	Marking Guidelines
P3	(a) $2^{2x+1} = 32$ $2^{2x+1} = 2^5$ $2x+1 = 5$ $x = 2$	2 marks correct solution 1 mark error made
P3	(b) $ x-2 \leq 3$ $-3 \leq x-2 \leq 3$ $-1 \leq x \leq 5$	2 marks correct solution 1 mark error made
P3 P4	(c) $36 - x^2 = 6^2 - x^2$ $= (6-x)(x+x)$	2 marks correct solution 1 mark error made
P3 P4	(d) $\text{y-coordinate} = \frac{-2+4}{2} = 1$ focus (5,1)	2 marks correct solution 1 mark error made 2 marks correct solution 1 mark error made

P3 P4	(e) (i) $\frac{1}{\sqrt{n} + \sqrt{n+1}} = \frac{1}{\sqrt{n} + \sqrt{n+1}} \times \frac{\sqrt{n} - \sqrt{n+1}}{\sqrt{n} - \sqrt{n+1}}$ $= \frac{\sqrt{n} - \sqrt{n+1}}{n - (n+1)}$ $= \frac{\sqrt{n} - \sqrt{n+1}}{-1}$ $= -\sqrt{n} + \sqrt{n+1}$	2 marks correct solution 1 mark error made
P3 P4	(ii) $\frac{1}{\sqrt{1} + \sqrt{2}} + \frac{1}{\sqrt{2} + \sqrt{3}} + \dots + \frac{1}{\sqrt{99} + \sqrt{100}} = \sqrt{2} - \sqrt{1} + \sqrt{3} - \sqrt{2} + \dots + \sqrt{100} - \sqrt{99}$ $= \sqrt{100} - \sqrt{1}$ $= 9$	3 marks correct solution 2 marks error made 1 mark establishes correct relationship of distances
P3 P4	(f) $PA^2 + PB^2 = 40$ $(x - (1))^2 + (y - 0)^2 + (x - 3)^2 + (y - 0)^2 = 40$ $x^2 + 2x + 1 + y^2 + x^2 - 6x + 9 + y^2 = 40$ $2x^2 + 2y^2 - 4x = 30$ $x^2 + y^2 - 2x = 15$ $x^2 - 2x + 1 + y^2 = 15 + 1$ $(x - 1)^2 + y^2 = 16$ Radius = 4 Centre (1, 0)	

Year 12 Mathematics Trial 2019		
Question No. 13		Solutions and Marking Guidelines
Outcomes Addressed in this Question		
P4 - chooses and applies appropriate arithmetic, algebraic, graphical and geometric techniques P5 - understands the concept of a function and the relationship between a function and its graph H6 - uses the derivative to determine the features of the graph of a function P6 - relates the derivative of a function to the slope of its graph P7 - determines the derivative of a function through routine application of the rules of differentiation H7 - uses the features of a graph to deduce information about the derivative H8 - uses techniques of integration to calculate areas and volumes		
Outcome	Solutions	Marking Guidelines
P4	(a) $\lim_{x \rightarrow 2} \frac{x^3 - 8}{x - 2} = \lim_{x \rightarrow 2} \frac{(x - 2)(x^2 + 2x + 4)}{x - 2}$ $= \lim_{x \rightarrow 2} x^2 + 2x + 4$ $= 12$	2 marks correct solution 1 mark error made

P7	(b) $y = \frac{x-1}{2x^2}$ $y' = \frac{3x^2(1) - 6x(x-1)}{(3x^2)^2}$ $= \frac{6x - 3x^2}{9x^2}$ $= \frac{2-x}{3x^3}$	2 marks correct solution 1 mark error made
H8	(c) $V = \pi \int_0^4 y^2 dx$ $= \pi \int_0^4 (1 + \sqrt{x})^2 dx$ $= \pi \int_0^4 1 + 2x^{\frac{1}{2}} + x dx$ $= \pi \left[x + \frac{2x^{\frac{3}{2}}}{\frac{3}{2}} + \frac{x^2}{2} \right]_0^4$ $= \pi \left(4 + \frac{4}{3} \times 4 \times 2 + 8 - 0 \right)$ $= \frac{68\pi}{3} \text{ units}^3$	3 marks correct solution 2 marks error made 1 mark demonstrates some relevant knowledge about volumes and integration
H6	(d) (i) $y = x^3 - x^2 - x + 1$ $y' = 3x^2 - 2x - 1$ $0 = 3x^2 - 2x - 1$ $0 = (3x+1)(x-1)$ $x = \frac{-1}{3}, 1$ At $\left(-\frac{1}{3}, \frac{32}{27}\right)$ $y'' = 6\left(-\frac{1}{3}\right) - 2 = -4 < 0$ Therefore maximum At $(1, 0)$ $y'' = 6(1) - 2 = 4 > 0$ Therefore minimum	4 marks correct solution 3 marks error made 2 marks finds correct x-values of stationary points 1 mark demonstrates some relevant knowledge about using calculus to find stationary points

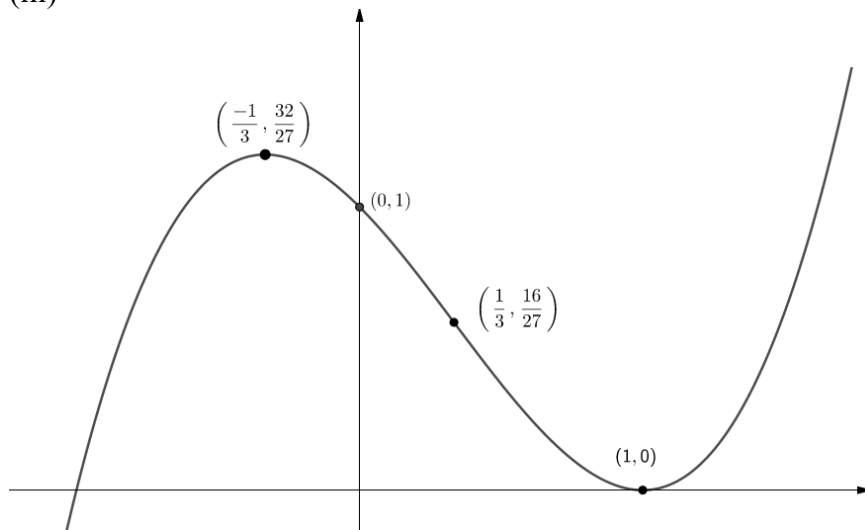
P6
H7

(ii)

x	0	$\frac{1}{3}$	$\frac{1}{2}$
y''	-2	0	1

There is a change in concavity at $x = \frac{1}{3}$, since there is a change in sign. Therefore $\left(\frac{1}{3}, \frac{16}{27}\right)$ is a point of inflection

(iii)



H6
H9

2 marks
correct solution with
concavity checked
1 mark
error made

2 marks
correct graph,
including curvature
and y-intercept
1 mark
error made

Year 12 Question 13	Mathematics Solutions and Marking Guidelines	Trial HSC (Task 4) 2019
Outcome Addressed in this Question		
H5 applies appropriate techniques from the study of calculus, geometry, probability, trigonometry and series to solve problems		
Part	Solutions	Marking Guidelines
(a)	$\tan x = \frac{1}{3}$ $x = 0.32 \text{ or } 3.46 \text{ (to 2 d.p.)}$	Award 2 for correct solutions Award 1 for substantial progress towards solution or answers in degrees i.e. $x = 18.43^\circ, 198.43^\circ$ or $18^\circ 26', 198^\circ 26'$
(b)	Length of arc $PQ = 10 \times \frac{2\pi}{3} = \frac{20\pi}{3}$ $\text{Length } TQ = \sqrt{5^2 + 10^2 - 2 \times 5 \times 10 \times \cos\left(\frac{2\pi}{3}\right)}$ $= \sqrt{25 + 100 - 100 \times \left(-\frac{1}{2}\right)}$ $= \sqrt{125 + 50}$ $= \sqrt{175} = \sqrt{25 \times 7} = 5\sqrt{7}$ Perimeter $PTQ = \text{length of arc } PQ + \text{length } PT + \text{length } TQ$ $= \left(\frac{20\pi}{3} + 5 + 5\sqrt{7}\right) \text{ cm}$	Award 3 for correct solution Award 2 for substantial progress towards solution Award 1 for limited progress towards solution
(c)	$A = \int_{\frac{\pi}{3}}^{\pi} (\sin x - \sin 2x) dx$ $= \left[-\cos x + \frac{1}{2} \cos 2x \right]_{\frac{\pi}{3}}^{\pi}$ $= \left[-\cos \pi + \frac{1}{2} \cos 2\pi \right] - \left[-\cos \frac{\pi}{3} + \frac{1}{2} \cos 2\left(\frac{\pi}{3}\right) \right]$ $= \left[1 + \frac{1}{2} \right] - \left[-\frac{1}{2} + \frac{1}{2} \left(-\frac{1}{2}\right) \right]$ $= 1 + \frac{1}{2} + \frac{1}{2} + \frac{1}{4} = 2\frac{1}{4} \text{ square units}$	Award 3 for correct solution Award 2 for substantial progress towards solution Award 1 for limited progress towards solution
(d) (i)	$L = \text{length } XY = XZ + YZ$ $XZ = \frac{1}{\cos \theta}$ $YZ = \frac{27}{\sin \theta}$ $\therefore L = \frac{1}{\cos \theta} + \frac{27}{\sin \theta}$	Award 1 for correct solution

(ii)

$$\begin{aligned}
 L &= \frac{1}{\cos \theta} + \frac{27}{\sin \theta} \\
 \frac{dL}{d\theta} &= \frac{-1(-\sin \theta)}{\cos^2 \theta} + \frac{-27 \cos \theta}{\sin^2 \theta} \\
 &= \frac{\sin \theta}{\cos^2 \theta} - \frac{27 \cos \theta}{\sin^2 \theta} \\
 &= \frac{\sin^3 \theta - 27 \cos^3 \theta}{\cos^2 \theta \sin^2 \theta} \\
 &= \frac{1}{\sin^2 \theta} \left(\frac{\sin^3 \theta - 27 \cos^3 \theta}{\cos^2 \theta} \right) \\
 &= \frac{1}{\sin^2 \theta} \times \frac{\cos \theta}{\cos \theta} \left(\frac{\sin^3 \theta - 27 \cos^3 \theta}{\cos^2 \theta} \right) \\
 &= \frac{\cos \theta}{\sin^2 \theta} \left(\frac{\sin^3 \theta - 27 \cos^3 \theta}{\cos^3 \theta} \right) \\
 &= \frac{\cos \theta}{\sin^2 \theta} \left(\frac{\sin^3 \theta}{\cos^3 \theta} - \frac{27 \cos^3 \theta}{\cos^3 \theta} \right) \\
 \therefore \frac{dL}{d\theta} &= \frac{\cos \theta}{\sin^2 \theta} (\tan^3 \theta - 27)
 \end{aligned}$$

Award 3 for correct solution

Award 2 for substantial progress towards solution

Award 1 for limited progress towards solution

(iii)

$$\frac{dL}{d\theta} = \frac{\cos \theta}{\sin^2 \theta} (\tan^3 \theta - 27)$$

$$\text{for minimum value } \frac{dL}{d\theta} = 0$$

$$\frac{\cos \theta}{\sin^2 \theta} (\tan^3 \theta - 27) = 0$$

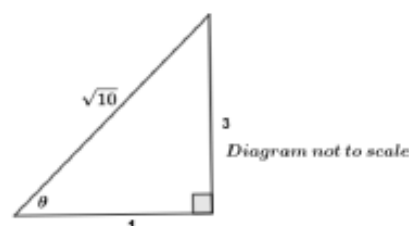
$$\frac{\cos \theta}{\sin^2 \theta} = 0 \quad \text{or} \quad (\tan^3 \theta - 27) = 0$$

$$\cos \theta = 0$$

$$\text{No solution as } 0 < \theta < \frac{\pi}{2}$$

$$\tan^3 \theta = 27$$

$$\tan \theta = 3$$



Award 3 for correct solution

Award 2 for substantial progress towards solution

Award 1 for limited progress towards solution

$$\sin \theta = \frac{3}{\sqrt{10}} \text{ and } \cos \theta = \frac{1}{\sqrt{10}}$$

$$L = \frac{1}{\frac{1}{\sqrt{10}}} + \frac{27}{\frac{3}{\sqrt{10}}}$$

$$L = \sqrt{10} \left(\frac{1}{1} + \frac{27}{3} \right)$$

$$L = \sqrt{10} \left(\frac{30}{3} \right)$$

$$\therefore L = 10\sqrt{10} \text{ units}$$

Year 12 Mathematics Trial 2019

Question No. 14

Solutions and Marking Guidelines

Outcomes Addressed in this Question

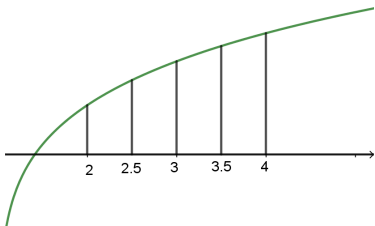
H3 Manipulates algebraic expressions involving logarithmic & exponential functions

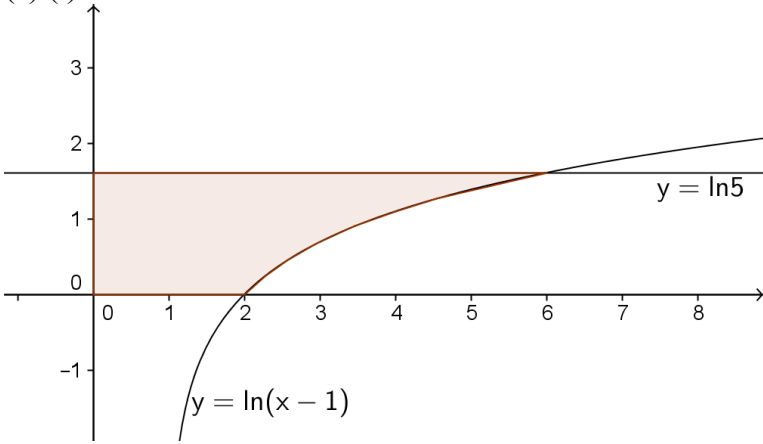
H5 Applies appropriate techniques from the study of calculus, geometry, probability, trigonometry & series to solve problems

H8 Uses techniques of integration to calculate areas & volumes

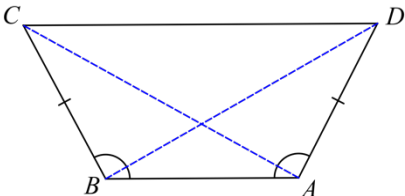
P5 Understands the concept of a function and the relationship between a function and its graph

P7 Determines the derivative of a function through routine application of the rules of differentiation

Outcome	Solutions	Marking Guidelines
H3	$(a) \log_5 8 = \log_5 k - 2 \log_5 3$ $\therefore \log_5 8 = \log_5 k - \log_5 3^2$ $\therefore \log_5 8 = \log_5 \frac{k}{9}$ $\therefore 8 = \frac{k}{9}$ $k = 72$	2 marks: correct solution 1 mark: substantial progress towards correct solution
H5	(b)  $f(x) = \ln(x^2 - 1)$ $A = \frac{0.5}{2} \{f(2) + 2[f(2.5) + f(3) + f(3.5)] + f(4)\}$ $= \frac{0.5}{2} \{\ln 3 + 2[\ln 5.25 + \ln 8 + \ln 11.25] + \ln 15\}$ $= 4.03 \text{ (to 2 decimal places)}$	3 marks: correct solution 2 marks: substantially correct solution 1 mark: substantial progress towards correct solution
H8	(c) $\int \frac{1}{2x-1} dx = \frac{1}{2} \int \frac{2}{2x-1} dx$ $= \frac{1}{2} \ln 2x-1 + c$	2 marks: correct solution 1 mark: substantial progress towards correct solution

H3, H5	(d) $f(x) = \log_e \sqrt{2-x}$ $= \log_e (2-x)^{\frac{1}{2}}$ $= \frac{1}{2} \log_e (2-x)$ $f'(x) = \frac{1}{2} \cdot \frac{-1}{2-x}$ $= \frac{-1}{2(2-x)}$	2 marks: correct solution 1 mark: substantial progress towards correct solution
P5, H5	(e) (i) 	1 mark: correct solution
H3, H8	(ii) Shaded area $= \int_0^{\ln 5} x \, dy$ If $y = \log_e (x-1)$, $e^y = x-1$ $\therefore x = e^y + 1$ $\therefore \text{shaded area} = \int_0^{\ln 5} (e^y + 1) \, dy$ $= [e^y + y]_0^{\ln 5}$ $= e^{\ln 5} + \ln 5 - (e^0 + 0)$ $= 5 + \ln 5 - 1$ $= 4 + \ln 5 \, \text{u}^2$	3 marks: correct solution 2 marks: substantially correct solution 1 mark: substantial progress towards correct solution
P7, H5	(f) (i) $\frac{d}{dx}(xe^x) = x \frac{d}{dx}(e^x) + e^x \frac{d}{dx}(x)$ (Product rule) $= xe^x + e^x \cdot 1$ $\therefore \frac{d}{dx}(xe^x) = xe^x + e^x$ Note: as question said 'show that' evidence was needed of How the answer was reached. This needed to be specific to this question, such as what your u and v were when using the product rule	1 mark: correct solution 1 mark: correct solution

H5	(ii) Since $\frac{d}{dx}(xe^x) = xe^x + e^x$ $\int (xe^x + e^x) dx = xe^x + c$ $\therefore \int xe^x dx + \int e^x dx = xe^x + c$ $\therefore \int xe^x dx + e^x = xe^x + c$ $\therefore \int xe^x dx = xe^x - e^x + c$	
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Year 12 Question 15	Mathematics Solutions and Marking Guidelines	Trial HSC (Task 4) 2019
Outcome Addressed in this Question P2 - provides reasoning to support conclusions which are appropriate to the context H2 - constructs arguments to prove and justify results H5 - applies appropriate techniques from the study of calculus, geometry, probability, trigonometry and series to solve problems H9 - communicates using mathematical language, notation, diagrams and graphs		
Part	Solutions	Marking Guidelines
(a) (i) P2, H2, H5	 Consider $\triangle ABC$ and $\triangle BAD$ $BC = AD$ (given) $\angle ABC = \angle BAD$ (given) $AB = AB$ (common side) $\triangle ABC \equiv \triangle BAD$ (SAS)	Award 2 for correct solution Award 1 for substantial progress towards solution
(ii) P2, H2	$\angle CAB = \angle ABD$ (matching angles in congruent triangles)	Award 1 for correct solution
(iii) H2, H5	$\angle ABC = \angle BAD$ $\angle DBC + \angle ABD = \angle CAD + \angle CAB$ (adjacent angles) Since $\angle CAB = \angle ABD$ (from part (ii)) Therefore $\angle DBC = \angle CAD$	Award 2 for correct solution Award 1 for substantial progress towards solution
(b) (i) H5	$x = \frac{1+3}{2}, y = \frac{-3+3}{2}$ $L = (2, 0)$ $x = \frac{3-3}{2}, y = \frac{3+1}{2}$ $M = (0, 2)$	Award 1 for correct solutions

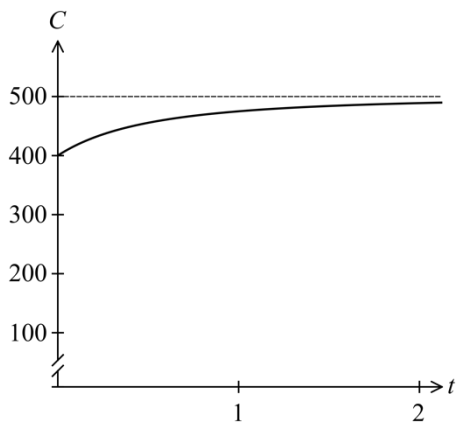
(ii) H2, H5	$m_{LM} = \frac{2 - 0}{0 - 2}$$= -1 \qquad \text{b}$$m_{\overrightarrow{AC}} = \frac{1 + 3}{-3 - 1}$$= -1 \quad \therefore LM \parallel AC$$LM = \sqrt{(0 - 2)^2 + (2 - 0)^2}$$= \sqrt{4 + 4}$$= \sqrt{8}$$= 2\sqrt{2}$$\overrightarrow{AC} = \sqrt{(-3 - 1)^2 + (1 + 3)^2}$$= \sqrt{16 + 16}$$= \sqrt{32}$$= 4\sqrt{2}$$\therefore 2 \times LM = AC$$LM = \frac{1}{2} AC$	Award 2 for correct solution Award 1 for substantial progress towards solution
(c) (i) H5, H9	$\frac{1}{6}$ Petra $\frac{1}{8}$ A $\frac{1}{48}$ PA $\frac{1}{4}$ B $\frac{1}{24}$ PB $\frac{3}{8}$ C $\frac{3}{48}$ PC $\frac{1}{4}$ D $\frac{1}{24}$ PD $\frac{1}{3}$ Quentin $\frac{1}{8}$ A $\frac{1}{24}$ QA $\frac{1}{4}$ B $\frac{1}{12}$ QB $\frac{3}{8}$ C $\frac{1}{8}$ QC $\frac{1}{4}$ D $\frac{1}{12}$ QD $\frac{1}{2}$ Rufus $\frac{1}{8}$ A $\frac{1}{16}$ RA $\frac{1}{4}$ B $\frac{1}{8}$ RB $\frac{3}{8}$ C $\frac{3}{16}$ RC $\frac{1}{4}$ D $\frac{1}{8}$ RD	Award 2 for correct solution Award 1 for substantial progress towards solution

(ii) H5	Outcomes are those marked E on the diagram $P(Q \text{ or } B \text{ but not both}) = \frac{1}{24} + \frac{1}{24} + \frac{1}{8} + \frac{1}{12} + \frac{1}{8}$ $= \frac{10}{24} = \frac{5}{12}$	Award 1 for correct solution
(d) (i) H5 H1	$r = 6\% \text{ pa} = 0.5\% \text{ per month} = 0.005$ $M_1 = P(1.005)$ $M_2 = (P(1.005) + P) \times 1.005$ $= P(1.005)^2 + P(1.005)$ $= P(1.005^2 + 1.005)$	Award 1 for correct solution
(ii) H5	$M_{12} = P(1.005^{12} + 1.005^{11} + \dots + 1.005)$	Award 1 for correct solution
(iii) H5	$n = 12 \times 20 = 240 \quad M_{240} = \450000 $S_{240} = \frac{1.005(1.005^{240} - 1)}{1.005 - 1}$ $= 464.351$ $P = \frac{450000}{464.351}$ $= \$969.09$	Award 2 for correct solution Award 1 for substantial progress towards solution

Year 12 Mathematics Trial HSC (Task 4) 2019 Question 16 Solutions and Marking Guidelines		
Outcome Addressed in this Question		
H4 - Expresses practical problems in mathematical terms based on simple given models. H1 - seeks to apply mathematical technique to problems in a wide range of practical context		
Part	Solutions	Marking Guidelines
(a) (i)	$V = 3000e^{-kt}$ $\frac{dV}{dt} = -k \times 3000e^{-kt} = -kV$	Award 1 for correct solution
(ii)	When $t = 3$ then $V = 2000$	Award 2 for correct solution Award 1 for substantial progress towards solution

(iii)	$2000 = 3000e^{-3k}$ $e^{-3k} = \frac{2}{3}$ $\log_e e^{-3k} = \log_e 0.6$ $-3k = \log_e 0.6$ $k = \frac{\log_e 0.6}{-3} = 0.135155036... \approx 0.135$ $2000 = 3000e^{-3k}$ $e^{-3k} = \frac{2}{3}$ $\log_e e^{-3k} = \log_e 0.6$ $-3k = \log_e 0.6$ $k = \frac{\log_e 0.6}{-3} = 0.135155036... \approx 0.135$	Award 2 for correct solution Award 1 for substantial progress towards solution
(b) (i)	We need to find t when $V = 250$. $250 = 3000e^{-k \times t}$ $e^{-kt} = 0.08\dot{3}$ $-kt = \log_e 0.08\dot{3}$ $t = -\frac{1}{k} \log_e 0.08\dot{3} = -\frac{\log_e 0.08\dot{3}}{0.135155..}$ $= 18.385601.. \approx 18 \text{ h } 23 \text{ min}$	Award 1 for correct solution
(ii)	Initial calculation occurs on 1st January 2011 or $t = 0$ $C = 500 - \left(\frac{10}{1+0} \right)^2$ $= 400 \text{ tonnes per year}$	Award 1 for correct solution
(iii)	1st January 2013 requires $t = 2$ $C = 500 - \left(\frac{10}{1+2} \right)^2$ $= 488.8 \text{ tonnes per year}$	Award 1 for correct solution
(iv)	$C = \lim_{t \rightarrow \infty} 500 - \left(\frac{10}{1+t} \right)^2 \quad \left(\lim_{t \rightarrow \infty} \frac{10}{1+t} \approx 0 \right)$ $\approx 500 \text{ tonnes per year}$	Award 1 for correct solution

(v)



Award 2 for correct solution

Award 1 for substantial progress towards solution

(c) (i)

Area under the curve represents the amount of carbon pollution.

$$\begin{aligned}\int_0^6 500 - \left(\frac{10}{1+t}\right)^2 dt &= \int_0^6 500 - 100(1+t)^{-2} dt \\ &= \left[500t + 100(1+t)^{-1} \right]_0^6 \\ &= \left[(500 \times 6 + 100(1+6)^{-1}) - (100(1+0)^{-1}) \right] \\ &= 2914.285714... \\ &\approx 2914 \text{ tonnes}\end{aligned}$$

Award 1 for correct solution

(ii)

When $t = 5$,

$$3200 = 2000e^{k \times 5}$$

$$e^{5k} = \frac{3200}{2000}$$

Award 1 for correct solution

(iii)

$$5k = \log_e \frac{8}{5}$$

$$k = \frac{1}{5} \log_e \frac{8}{5}$$

$$= 0.09400072...$$

Award 1 for correct solution

We need to find M when $t = 10$

$$M = 2000e^{k \times 10}$$

$$= 5120$$

(iv)

We need to find t when $M = 4000$.

$$4000 = 2000e^{k \times t}$$

$$e^{kt} = 2$$

$$kt = \log_e 2$$

$$t = \frac{1}{k} \log_e 2$$

$$= 5 \frac{\log_e 2}{\log_e \frac{8}{5}}$$

$$= 7.373849237.. \approx 7.4 \text{ years}$$

$$M = 2000e^{kt}$$

$$\frac{dM}{dt} = k2000e^{kt}$$

$$= kM$$

$$= \frac{1}{5} \log_e \frac{8}{5} \times 3200$$

$$= 300.8023227... \approx 301$$

Award 1 for correct solution